

- 12 (a)** (carrier wave) transmitted from Earth to satellite (1)  
 satellite receives greatly attenuated signal (1)  
 signal amplified and transmitted back to Earth B1  
 at a different (carrier) frequency B1  
 different frequencies prevent swamping of uplink signal (1)  
 e.g. of frequencies used (6/4 GHz, 14/11 GHz, 30/20 GHz) (1)  
*(two B1 marks plus any two other for additional physics)* B2 [4]
- (b)** advantage: e.g. much shorter time delay M1  
 because orbits are much lower A1  
 e.g. whole Earth may be covered (M1)  
 in several orbits / with network (A1)
- disadvantage: e.g. *either* must be tracked M1  
*or* limited use in any one orbit A1  
 more satellites required for continuous operation [4]